

《复变函数与积分变换》2022年答案

1. $e^{2-3i} = e^2 (\cos 3 - i \sin 3) = \boxed{e^2 \cos 3 - i e^2 \sin 3}$

$\rightarrow \arg z = \frac{\pi}{3}$

2. 计算 $(1+\sqrt{3}i)^{1-i} = e^{(1-i) \ln(1+\sqrt{3}i)}$

$= e^{(1-i) [\ln 2 + i(\frac{\pi}{3} + 2k\pi)]}$

$= e^{\ln 2 + \frac{\pi}{3} + 2k\pi + i(\frac{\pi}{3} + 2k\pi - \ln 2)}$

$= e^{\ln 2 + \frac{\pi}{3} + 2k\pi} [\cos(\frac{\pi}{3} + 2k\pi - \ln 2) + i \sin(\frac{\pi}{3} + 2k\pi - \ln 2)]$

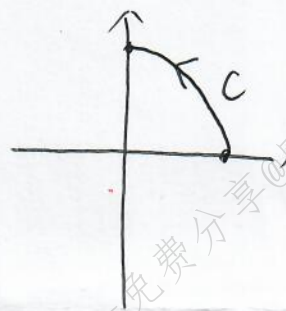
$= \boxed{2 e^{\frac{\pi}{3} + 2k\pi} [\cos(\frac{\pi}{3} - \ln 2) + i \sin(\frac{\pi}{3} - \ln 2)]}, k \in \mathbb{Z}$

3. $\int_C (\bar{z} + z) dz$, 其中C为从1到i沿 $|z|=1$ 的一段弧。

解: $= \int_C (\frac{|z|^2}{z} + z) dz$

解利 $= \int_1^i (\frac{1}{z} + z) dz$

$= (\ln z + \frac{z^2}{2}) \Big|_1^i = \ln i + \frac{i^2}{2} - \frac{1}{2} = \boxed{\frac{\pi}{2}i - 1}$



4. 设 $f(z) = \frac{1}{z(z+1)}$, 则 $f^{(6)}(2) =$

解: 由高阶导数公式 $f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^{n+1}} dz$, 其中C为包围 z_0 的闭曲线

$\therefore f^{(6)}(2) = \frac{6!}{2\pi i} \oint_{|z-2|=\gamma} \frac{1}{z(z+1)} dz$

$= \frac{6!}{2\pi i} \oint_{|z-2|=\gamma} \frac{1}{z(z+1)(z-2)^7} dz$

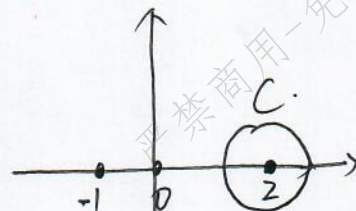
留数定理

$= \frac{6!}{2\pi i} \times 2\pi i \operatorname{Res}[\frac{1}{z(z+1)(z-2)^7}, 2]$

留数第二定理 $= -6! \times \{ \operatorname{Res}[\frac{1}{z(z+1)(z-2)^7}, 0] + \operatorname{Res}[\frac{1}{z(z+1)(z-2)^7}, -1] + \operatorname{Res}[\frac{1}{z(z+1)(z-2)^7}, \infty] \}$

$= -6! \times \{ \frac{1}{-2^7} + \frac{1}{(-1) \times (-2)^7} - \operatorname{Res}[\frac{1}{\frac{1}{z} \cdot (\frac{1}{z+1}) (\frac{1}{z-2})^7}, 0] \}$

$= \boxed{6! \times (\frac{1}{-2^7} - \frac{1}{2^7})}$



$\star |z-2|=\gamma$ 不含0和-1

$$5. \int_{-\infty}^{+\infty} \cos(2t) e^{-3t} dt \xrightarrow{\text{收敛}} \int_0^{+\infty} \cos(2t) e^{-3t} dt$$

(题目打印错误)

$$= \mathcal{L}[\cos(2t)]|_{s=3}$$

$$= -\left(\frac{s}{s^2+4}\right)' \Big|_{s=3} = \frac{s^2-4}{(s^2+4)^2} \Big|_{s=3} = \boxed{\frac{5}{169}}$$

$$二. f(z) = \frac{\sin \pi z}{(z^2-1)(e^{i\pi z}+1)} \text{ 在扩充复平面内的孤立奇点及分类}$$

$$\text{解: 令 } p(z) = \sin \pi z, \quad q(z) = (z^2-1)(e^{i\pi z}+1)$$

$$f(z) \text{ 的奇点有: } z_k = 2k+1 \quad (k=0, \pm 1, \dots), \quad z_\infty = \infty \quad (\text{非孤立})$$

$$(q(z) \rightarrow 0) \quad \downarrow \quad \text{孤立奇点}$$

$$\lim_{k \rightarrow \infty} z_k = +\infty \text{ 在 } |z| > R \text{ 内}$$

$$\textcircled{1} z = \pm 1, \text{ 为 } p(z) \text{ 的 1 级零点, } q(z) \text{ 的 2 级零点}$$

$$\Rightarrow \text{为 } f(z) \text{ 的 1 级极点}$$

$$\textcircled{2} z = 2k+1 \quad (k=0, \pm 1, \dots), \text{ 为 } p(z) \text{ 的 1 级零点, } q(z) \text{ 的 1 级零点} \Rightarrow \text{为 } f(z) \text{ 的可去奇点}$$

$$\text{综上: } z = \pm 1 \text{ 为 1 级极点, } z_k = 2k+1 \quad (k \neq 0, \pm 1) \text{ 为可去奇点}$$

$$三. 1. \oint_{|z|=3} \frac{1}{(z+1)^3(z-2)^2(z+3)} dz \xrightarrow{f(z)}$$

$$= -2\pi i \{ \text{Res}[f(z), -3] + \text{Res}[f(z), \infty] \}$$

$$= -2\pi i \times \left\{ \frac{1}{10^3 \times (-5)^2} - \text{Res}\left[\frac{1}{\left(\frac{1}{3}+1\right)^3 \left(\frac{1}{2}-2\right)^2 \left(\frac{1}{2}+3\right)}, \frac{1}{2}, 0\right] \right\}$$

$$= \boxed{-\frac{\pi i}{12500}}$$

$$2. \oint_{|z|=3} \frac{z}{(e^z-1)(z^2+4)} dz$$

$$= 2\pi i \times \{ \text{Res}[f(z), 0] + \text{Res}[f(z), 2i] + \text{Res}[f(z), -2i] \}$$

$$= 2\pi i \times \left\{ 0 + \frac{2i}{[e^{2i}-1] \times 4i} + \frac{-2i}{[e^{-2i}-1] \times (-4i)} \right\}$$

$$= \frac{2\pi i (\cos 2 - 1)}{(\cos 2 - 1)^2 + \sin^2 2} = \frac{2\pi i (\cos 2 - 1)}{2(1 - \cos 2)}$$

$$= \boxed{-\pi i}$$



$$3. \int_0^{+\infty} \frac{e^{-2t} - e^{-3t}}{t} dt$$

$$= \int_0^{+\infty} \frac{e^{-2t}}{t} dt - \int_0^{+\infty} \frac{e^{-3t}}{t} dt$$

$$= \int_2^{\infty} \frac{1}{s} ds - \int_3^{\infty} \frac{1}{s} ds$$

$$= \int_2^{\infty} \frac{1}{s} ds - \int_3^{\infty} \frac{1}{s} ds = \int_2^3 \frac{1}{s} ds$$

$$= \ln s \Big|_2^3$$

$$= \ln 3 - \ln 2$$

$$5. \int_0^{2\pi} \frac{1}{3+2\sin\theta} d\theta$$

$$= \oint_{|z|=1} \frac{1}{3+2\frac{z-\frac{1}{z}}{2i}} \cdot \frac{1}{iz} dz$$

$$= \oint_{|z|=1} \frac{1}{z^2+3iz-1} dz = 2\pi i \times \text{Res} \left[\frac{1}{z^2+3iz-1}, \frac{-3+\sqrt{5}}{2} i \right] = \frac{2\sqrt{5}\pi}{5}$$

IV. $f(z) = \frac{1}{z^2+1}$ 在以 $z_0 = 2i$ 为中心的圆环域内展开成洛朗级数.

解: 圆环域有: $1 < |z-2i| < 3$ 和 $3 < |z-2i| < +\infty$

(注意: $|z-2i| < 1$ 不是圆环域, 因为 $z=2i$ 是解折点)

是圆域, 相应的展开和泰勒展开 (分不消和直接展开都可以)

$$f(z) = \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right)$$

$$i) \underline{1 < |z-2i| < 3}, \quad \frac{1}{z-i} = \frac{1}{i+z-2i} = \frac{1}{z-2i} \cdot \frac{1}{1+\frac{i}{z-2i}} = \frac{1}{z-2i} \sum_{n=0}^{\infty} (-1)^n \left(\frac{i}{z-2i} \right)^n$$

$$\frac{1}{z+i} = \frac{1}{3i+z-2i} = \frac{1}{3i} \cdot \frac{1}{1+\frac{z-2i}{3i}} = \frac{1}{3i} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-2i}{3i} \right)^n$$

$$\therefore f(z) = \frac{1}{2i} \left[\sum_{n=0}^{\infty} (-1)^n \frac{i^n}{(z-2i)^{n+1}} - \sum_{n=0}^{\infty} (-1)^n \frac{(z-2i)^n}{(3i)^{n+1}} \right]$$

$$ii) \underline{3 < |z-2i| < +\infty}, \quad \frac{1}{z+i} = \frac{1}{3i+z-2i} = \frac{1}{z-2i} \cdot \frac{1}{1+\frac{3i}{z-2i}} = \frac{1}{z-2i} \sum_{n=0}^{\infty} (-1)^n \left(\frac{3i}{z-2i} \right)^n$$

$$\therefore f(z) = \frac{1}{2i} \left[\sum_{n=0}^{\infty} (-1)^n \frac{i^n}{(z-2i)^{n+1}} - \sum_{n=0}^{\infty} (-1)^n \frac{(3i)^n}{(z-2i)^{n+1}} \right]$$

$$4. \int_{-\infty}^{+\infty} \frac{x \cos x}{x^2-2x+4} dx. \quad \text{5.3节 (第三类)}$$

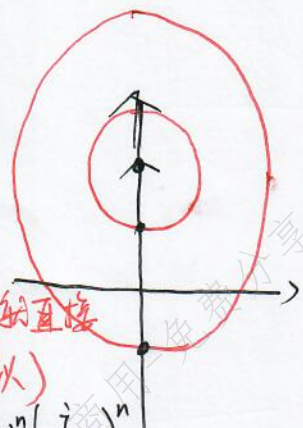
$$= \text{Re} \left[\int_{-\infty}^{+\infty} \frac{x e^{ix}}{x^2-2x+4} dx \right]$$

$$= \text{Re} \left(2\pi i \times \text{Res} \left[\frac{z e^{iz}}{z^2-2z+4}, 1+\sqrt{3}i \right] \right)$$

$$= \text{Re} \left(2\pi i \times \lim_{z \rightarrow 1+\sqrt{3}i} \frac{z e^{iz}}{2z-2} \right)$$

$$= \text{Re} \left(2\pi i \times \frac{(1+\sqrt{3}i) e^{i(1+\sqrt{3}i)}}{2(1+\sqrt{3}i-1)} \right)$$

$$= \frac{\pi}{\sqrt{3}e^{\sqrt{3}}} (\cos 1 - \sqrt{3} \sin 1)$$



五. 设 $u = xy + y - 2$, 证明其为调和函数. 并求 v 使得 $f(z) = u + iv$ 解析并满足 $f(1+i) = 0$

解: i) $\frac{\partial u}{\partial x} = y, \quad \frac{\partial u}{\partial y} = x+1.$
 $\frac{\partial^2 u}{\partial x^2} = 0, \quad \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0+0=0 \Rightarrow u$ 为调和函数

ii) 由 C-R 方程

$$\begin{cases} \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = y & ① \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -(x+1) & ② \end{cases}$$

由 ① $\Rightarrow v = \int y dy = \frac{y^2}{2} + \phi(x) \quad \therefore \frac{\partial v}{\partial x} = \phi'(x)$

由 ② $\Rightarrow \phi'(x) = -(x+1) \quad \therefore \phi(x) = -\frac{x^2}{2} - x + C$

$\therefore v(x, y) = \frac{y^2}{2} - \frac{x^2}{2} - x + C \Rightarrow f(z) = u + iv = xy + y - 2 + i(\frac{y^2}{2} - \frac{x^2}{2} - x + C)$

由 $f(1+i) = 0 \Rightarrow$ 当 $x=1, y=1$ 时. $u+iv=0$

$\Rightarrow 1+1-2 + i(\frac{1}{2} - \frac{1}{2} - 1 + C) = 0 \Rightarrow C = 1$

$\therefore f(z) = xy + y - 2 + i(\frac{y^2}{2} - \frac{x^2}{2} - x + 1)$

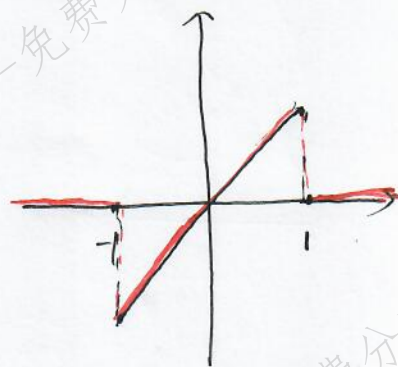
六. 求 $f(t) = \begin{cases} t, & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$ 的傅氏变换, 并计算积分 $\int_0^{+\infty} \frac{(\sin w - w \cos w) \sin wt}{w^2} dw$.

解: i) $F(w) = \int_{-\infty}^{+\infty} f(t) e^{-iwt} dt = \int_{-1}^1 t \cdot e^{-iwt} dt$

$= -\frac{1}{i w} \left[t e^{-iwt} \Big|_{-1}^1 - \int_{-1}^1 e^{-iwt} dt \right]$

$= -\frac{1}{i w} \left[e^{-iwt} + e^{iwt} - \frac{e^{-iwt}}{-i w} \Big|_{-1}^1 \right]$

$= -\frac{1}{i w} \cdot 2 \left(\cos w - \frac{\sin w}{w} \right) = \frac{2(\sin w - w \cos w)}{i w^2}$



ii) $\therefore f^*(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(w) e^{iwt} dw = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{2(\sin w - w \cos w)}{i w^2} \cdot (\cos wt + i \sin wt) dw$

$= \frac{2}{\pi} \int_0^{+\infty} \frac{(\sin w - w \cos w) \sin wt}{w^2} dw = \begin{cases} 0, & |t| > 1 \\ t, & |t| < 1 \end{cases}$

$\Rightarrow \int_0^{+\infty} \frac{(\sin w - w \cos w) \sin wt}{w^2} dw = \begin{cases} 0, & |t| > 1 \\ \frac{\pi t}{2}, & |t| < 1 \\ -\frac{\pi}{4}, & t = -1 \\ \frac{\pi}{4}, & t = 1 \end{cases}$

七. 求边值问题
$$\begin{cases} y'' + y = 10 \sin 2t \\ y(0) = 0, y(\frac{\pi}{2}) = 1. \end{cases}$$

解: 设 $z(y(t)) = Y(s)$

则对 $y'' + y = 10 \sin 2t$. 两边同时做 Laplace 变换.

$$\Rightarrow \mathcal{L}[y'' + y] = 10 \mathcal{L}[\sin 2t]$$

$$\Rightarrow s^2 Y(s) - s \overset{1,0}{y(0)} - y'(0) + Y(s) = 10 \times \frac{2}{s^2 + 4}$$

$$\Rightarrow \underline{Y(s)} = \frac{\frac{20}{s^2 + 4} + y'(0)}{s^2 + 1} = \frac{20}{(s^2 + 4)(s^2 + 1)} + \frac{y'(0)}{s^2 + 1}$$

$$\therefore y(t) = \mathcal{L}^{-1}[Y(s)]$$

$$= 20 \mathcal{L}^{-1}\left[\frac{1}{3} \left(\frac{1}{s^2 + 1} - \frac{1}{s^2 + 4}\right)\right] + y'(0) \mathcal{L}^{-1}\left[\frac{1}{s^2 + 1}\right]$$

$$= \frac{20}{3} \left(\sin t - \frac{\sin 2t}{2}\right) + y'(0) \sin t$$

将 $y(\frac{\pi}{2}) = 1$ 代入 $\Rightarrow y(\frac{\pi}{2}) = \frac{20}{3} \left(\sin \frac{\pi}{2} - \frac{\sin \pi}{2}\right) + y'(0) \sin \frac{\pi}{2}$

$$= \frac{20}{3} + y'(0) = 1$$

$$\Rightarrow \underline{y'(0)} = 1 - \frac{20}{3} = \underline{-\frac{17}{3}}$$

$$\therefore y(t) = \frac{20}{3} \left(\sin t - \frac{\sin 2t}{2}\right) + \left(-\frac{17}{3}\right) \sin t$$

$$= \boxed{\sin t - \frac{10}{3} \sin 2t}, \quad 0 \leq t \leq \frac{\pi}{2}$$